



Residual-Inception Siamese Architecture For Human Identification Via Palatal Rugae Ridge

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Abstract : Biometric identification has evolved significantly over the past decade, yet many conventional modalities such as fingerprints, facial recognition, and iris scanning remain vulnerable to environmental distortions, injuries, aging effects, or intentional alterations. To overcome these limitations, this project introduces a deep-learning-driven palatal rugae identification system, leveraging the anatomical stability and uniqueness of the rugae patterns located on the palate. The proposed approach integrates an enhanced ResNet-Inception-based Twin Encoder (Siamese) architecture capable of extracting fine-grained, multi-scale features from oral cavity images. The core of the system focuses on generating discriminative biometric embedding using a hybrid CNN architecture, combining both deep convolutional features and morphological characteristics of palatal ridges. During training, the Siamese network learns similarity and dissimilarity relationships using contrastive and triplet loss functions, enabling accurate verification even in the presence of age progression, illumination variations, and minor anatomical changes. A specialized embedding comparison mechanism and weighted similarity score (W_CNN) are used to match test images against stored templates through an efficient DP location-based retrieval process. Experimental results show that the proposed model significantly improves identification accuracy, reduces false matches, and ensures stability across multiple age groups. This makes the system highly suitable for forensic applications such as missing person identification, mass disaster victim identification, and dental record verification. The project demonstrates that combining advanced deep learning architectures with palatal rugae biometrics can establish a reliable, tamper-proof, and age-invariant identification framework for real-world forensic

IndexTerms - Palatal Rugae, Siamese Network, Deep Learning, Age progression, Human Identification, Residual Network

I. INTRODUCTION

Biometric identification has become an essential component of modern security, forensic investigation, and digital authentication systems. Traditional biometric traits such as fingerprints, iris patterns, and facial features have been widely used due to their uniqueness and ease of acquisition. However, these modalities can be influenced by external factors such as injuries, aging, environmental variations, or deliberate alteration. In many real-world forensic scenarios—including mass disaster victim identification, missing person investigations, and criminal profiling—these commonly used biometrics may be unavailable or unreliable. This practical limitation has led researchers to explore alternative physiological markers that remain stable over long periods of time. One such promising anatomical feature is the palatal rugae, the mucosal ridges found on the anterior part of the hard palate inside the oral cavity. Palatal rugae offer several inherent advantages that make them suitable for biometric identification. Their pattern is highly unique to each individual, remains stable throughout life, and is protected within the oral cavity from environmental damage or external manipulation. Furthermore, palatal rugae exhibit minimal changes even during aging, orthodontic treatment, or minor oral conditions, making them an ideal biometric trait for long-term study. Despite these advantages, manual comparison and identification based on palatal rugae has traditionally been challenging due to variations in illumination, tongue obstruction, saliva reflection, and inconsistent image capture angles. To overcome these issues and automate the identification process, advanced computational techniques and machine learning approaches have become essential.

In recent years, deep learning, particularly Convolutional Neural Networks (CNNs), has revolutionized the fields of pattern recognition and biometric identification. CNNs have the capability to automatically learn complex features from images without requiring predefined handcrafted descriptors. However, for tasks involving similarity comparison—such as determining whether two biometric samples belong to the same individual—standard CNN classification methods are insufficient. Instead, Siamese networks, which use twin neural architectures to learn similarity metrics between pairs of images, have become highly effective. Siamese networks generate embedding representing the identity of each biometric sample and measure their distance to determine similarity or dissimilarity. Building upon this foundation, the present project introduces an advanced ResNet-Inception based Twin Encoder Siamese Network specifically designed for palatal rugae identification. The hybrid architecture combines the strengths of two powerful deep learning models: ResNet, which enables very deep feature extraction without loss of information through residual learning; and Inception, which captures multi-scale features by processing images through parallel convolution kernels of different sizes. This dual capability allows the model to recognize subtle and complex structures present in palatal rugae patterns, such as ridge curvature, branching, length, and orientation, even under varying imaging conditions. The twin encoder design ensures

that both images in a pair undergo identical transformations and feature extraction, resulting in highly robust and comparable embedding.

To ensure the accuracy and stability of the system, the project employs a sophisticated preprocessing pipeline. This begins with the application of CLAHE (Contrast Limited Adaptive Histogram Equalization) to enhance local contrast and reveal hidden ridge details. Median filtering is used to remove salt-and-pepper noise and smooth regions without blurring edge structures. ROI extraction isolates the palatal region while removing irrelevant structures such as teeth, gums, or the tongue. Image normalization and resizing ensure that all samples are standardized in terms of intensity and spatial resolution before being passed to the neural network. This preprocessing pipeline significantly improves feature clarity and reduces the learning burden on the network. Another significant contribution of this project is the introduction of hybrid feature extraction, where deep CNN embeddings are combined with morphological descriptors derived from palatal ridge geometry. These morphological features—such as ridge count, length, curvature degree, and branching complexity—provide an additional layer of discriminatory power. The hybrid representation makes the identification system more robust, especially in cases where deep learning models alone may fail due to limited training data or extreme image variations.

During training, the system relies on contrastive loss and triplet loss to learn meaningful similarity metrics. These loss functions guide the Siamese network to bring embeddings of similar individuals closer while pushing embeddings of different individuals farther apart in multidimensional space. To handle real-world challenges such as age progression, this project introduces age-adaptive embedding stabilization, a mechanism that encourages the model to remain invariant to small morphological changes occurring naturally over time. Once the training is complete, the system stores all learned embeddings in a structured database. During testing, a palatal rugae image is preprocessed, encoded by the twin ResNet-Inception model, and compared against stored templates using a similarity distance metric. The system then determines the identity by selecting the nearest embedding, supported by a DP location mechanism for efficient and accurate retrieval.

Overall, this project contributes to the field of biometric identification by demonstrating that palatal rugae, combined with advanced deep learning architectures, provide a reliable, high-accuracy, and age-invariant biometric solution. The integration of multi-scale CNN feature extraction, morphological analysis, robust preprocessing techniques, and deep metric learning creates a comprehensive system suitable for forensic, security, and medical applications. This study establishes a strong foundation for future work in oral cavity biometrics and positions palatal rugae as a viable alternative in the landscape of next-generation biometric technologies.

I. PROBLEM STATEMENT

This paper introduces one of the earliest deep-learning-based frameworks for automatic human identification using palatal rugae images. The authors highlight that palatal rugae patterns behave as a unique biometric trait, remaining stable even after trauma or burns. The study uses high-resolution maxillary cast images as input, followed by preprocessing steps such as contrast enhancement, thresholding, and segmentation to isolate the rugae region. A Convolutional Neural Network (CNN) model is trained on the segmented images to learn pattern variations. The paper compares performance with classical methods like SIFT, SURF, and manual anthropometric measurements, showing CNN achieved significantly higher classification accuracy (above 90% on the dataset used). The study proves that deep features outperform handcrafted features in biometric recognition [1].

This paper focuses on classical digital image processing techniques to extract palatal rugae patterns for forensic identification. The workflow includes RGB to grayscale conversion, histogram equalization, thresholding, edge detection using Canny, and morphological operations to highlight rugae lines. The authors then apply geometric measurements such as rugae length, orientation, branching, shape classification (wavy, curved, straight), and matching using Euclidean distance. This approach is particularly useful because it explains the mathematical basis for matching—using coordinate points extracted along each rugae. This paper proposes a geometric modeling technique for palatal rugae using control points, spline curves, and Euclidean distance measures. The palatal area is segmented using thresholding, and each rugae is represented using a cubic spline. The extracted features include curvature, orientation, branching pattern, rugae depth, and a set of control points that uniquely represent the shape. The matching algorithm compares geometric vectors of two palatal patterns to determine identity [2][3].

This paper introduces a morphological pattern analysis system focusing on rugae shape types: straight, angular, wavy, curved, branched, and circular. Using morphological filters and skeletonization, the authors extract structural features such as bifurcation points, endpoints, thickness, and texture density. A rule-based classifier is used for identification [4]. This advanced study uses a deep CNN classifier, training it with several convolutional layers to extract spatial and textural features of palatal rugae. The authors use data augmentation and dropout to reduce overfitting. The method achieves high accuracy and is robust to minor noise and brightness changes. They emphasize the stability of rugae shape across age groups [5].

3. SYSTEM ANALYSIS

The proposed palatal rugae-based human identification architecture is designed as a multi-stage, highly robust pipeline capable of functioning even under challenging forensic conditions such as partial burns, severe facial disfigurement, decomposed remains, or missing fingerprints. The entire system begins with the acquisition of high-resolution palatal images using intra-oral cameras, dental scanners, or scanned dental casts. Since palatal structures are protected by the oral cavity, their patterns tend to remain intact even under extreme circumstances. During acquisition, careful attention is given to maintain uniformity, such as holding the camera at a fixed angle, ensuring proper light distribution using ring flash mechanisms, and retracting cheeks and

tongue to make the palatal region clearly visible. This standardized protocol minimizes distortions and greatly enhances the effectiveness of subsequent computational modules.

From the Fig:1 Once the raw images are obtained, they enter the preprocessing and normalization stage. The purpose of this phase is to eliminate noise, enhance clarity, and bring all palatal images into a consistent geometric orientation. The images are first converted to grayscale to highlight the texture of the rugae structures. Uneven lighting and color variations are corrected through histogram equalization, which redistributes pixel intensities to improve the visibility of the intricate ridges. To avoid losing structural information, bilateral filtering is applied to smoothen the image while simultaneously retaining edges—something that is extremely important, as rugae lines are often thin and require preservation during noise reduction. The system then detects key anatomical landmarks such as the incisive papilla and the mid-palatal line. These landmarks allow the image to be normalized using a rotation-translation transformation. The transformation is applied through the normalization function, where the matrix aligns each palate to a common axis. This ensures that all images have the same orientation, scale, and framing before segmentation.

After preprocessing, the image is forwarded to the U-Net-based segmentation module, which plays a central role in isolating the palatal rugae from surrounding oral tissues. U-Net is specifically chosen because it has proven highly effective in biomedical segmentation tasks, particularly those involving limited datasets. The architecture works by extracting hierarchical features through an encoder network, while simultaneously reconstructing a pixel-accurate segmentation mask through a decoder network that uses skip connections to preserve fine details. The segmentation output highlights only the rugae ridges, removing irrelevant regions such as the surrounding mucosa or palate walls. To achieve this, the training loss function combines Dice loss and binary cross-entropy, expressed as . This hybrid loss ensures that thin rugae lines, which may otherwise be lost during segmentation, are preserved with greater clarity and accuracy.

Once the segmentation mask is extracted, the system applies a specialized enhancement pipeline known as Data Augmentation 2.0. Unlike traditional augmentation methods, this approach incorporates biological realism into the transformational operations. Elastic deformation is used to simulate natural stretching of palatal tissues, controlled mathematically through the function . This creates subtle variations without distorting the fundamental rugae shape. Illumination adjustments replicate real-world imaging conditions encountered in forensics. Additional age-aware augmentations such as texture thinning, ridge fading, and smoothening simulate how palatal rugae patterns appear in older individuals. These advanced augmentation strategies significantly increase the diversity of the training dataset, enabling the system to generalize well even when presented with unusual or degraded samples.

The next major component is the Hybrid Feature Extraction Module, which combines the strengths of deep neural networks and mathematical morphology. The segmented image is fed into a convolutional neural network (CNN), which automatically extracts high-level deep features representing texture, orientation patterns, rugae depths, and inter-ridge variations. The output of the CNN, represented as , forms a high-dimensional embedding that characterizes the global and local patterns of the rugae.

Although CNN features are powerful, forensic odontologists frequently rely on precise morphological measurements, which is why handcrafted features are incorporated alongside deep learning features. These morphological features include rugae length, curvature, branching patterns, orientation angles, shape categories, and inter-rugae distances. For instance, the length of each rugae can be computed using the summation of Euclidean distances along its skeleton line:

The hybrid feature vector is then passed to the Enhanced Siamese Network (ESN), which is responsible for performing similarity assessment and identity verification. The ESN consists of two weight-shared subnetworks that process a pair of palatal images simultaneously. These subnetworks project each input into a common embedding space. The distance between the embeddings reflects how similar or dissimilar the two palatal patterns are. Traditional Siamese networks compute Euclidean or cosine distance, but the enhanced version incorporates margin-based contrastive loss that handles both intra-subject variability and inter-subject distinctiveness. The contrastive loss function is generally formulated as

A unique aspect of this architecture is the incorporation of an age-progression handling module. As individuals age, their palatal structures may undergo minute changes such as slight smoothening of ridges, reduction of sharpness, and mild morphological stretching. To address this, the system implements an age-compensated embedding transformation model that maps features from older samples into a younger reference domain. This ensures that even decades-old samples can be matched reliably with fresh samples.

Finally, the output similarity score from the ESN is passed to the decision module, where a threshold-based classifier determines whether the palatal rugae image belongs to the same individual or not. The threshold is optimized using evaluation metrics such as accuracy, precision, recall, F1-score, ROC-AUC, and equal error rate (EER). These metrics collectively ensure that the system is not only accurate but also balanced in terms of false accept and false reject rates, which is extremely important in forensic identification scenarios

$$W_Final = \alpha.W_CNN + \beta.W_length + \gamma.W_angle + \psi.W_curve \quad (1)$$

W_length = rugae length similarity

W_angle = orientation similarity

W_curve = curvature similarity

W_CNN = ResNet inception cosine similarity

Where $\alpha, \beta, \psi, \gamma$ = coefficients

3.1. Image Acquisition

Direct Image Capture: Using intraoral cameras, DSLR macro lenses, or mobile devices equipped with oral retractors.

Image Upload: Accepting pre-captured palatal images or scanned dental casts.

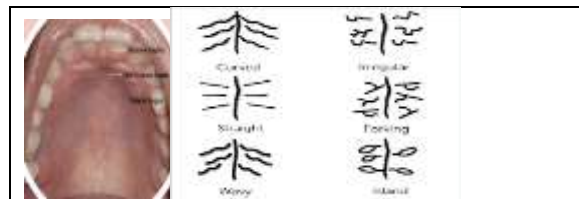


Figure 2: Various patterns of palatal rugae. Palatal rugae are classified by their shape, continuity, and direction. The classification along with the varying length makes each rugae universally unique yet still a viable tool for identification.

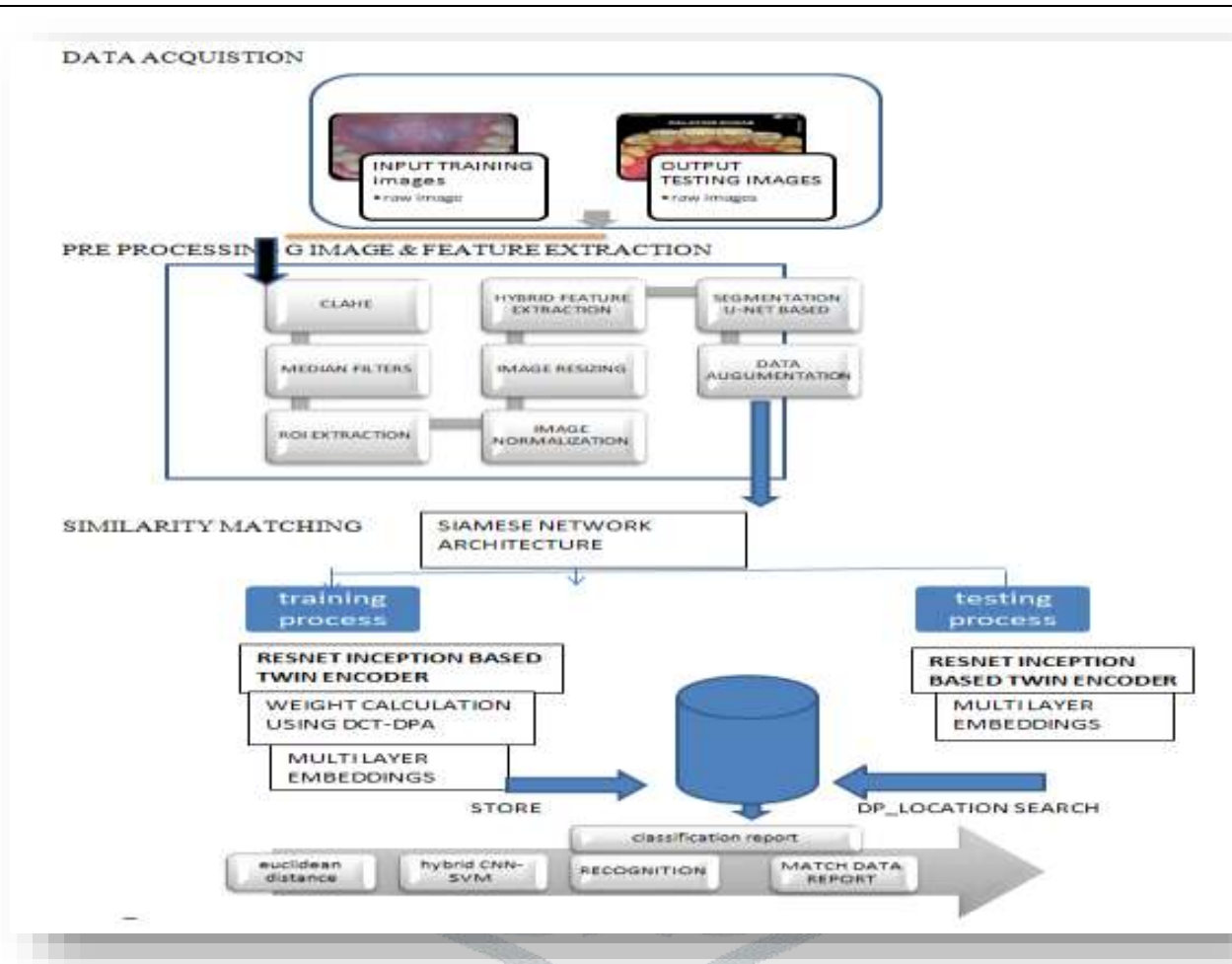


Figure 1: proposed system shows the working process of overall system with the Siamese network which clearly identifies the human with the palatal rugae structure with varied changes of age.

3.3. Hybrid Feature Extraction

A. CNN Feature Extraction

A deep CNN extracts high-dimensional embeddings from segmented palatal images. These embeddings encode: ridge texture, curvature, depth variations, fine structural differences, inter-ridge spacing

CNN features capture non-linear patterns that humans cannot easily measure.

B. Morphological Feature Extraction

These are handcrafted features derived using image processing and geometry. Examples:

1. Ridge Count: Number of primary and secondary rugae.
2. Ridge Orientation Angle: Average slope or angle of rugae.
3. Ridge Length: Using Euclidean distance formula.

4. Curvature Index: First and second derivative shape analysis.

5. Branching & Unification Features: Morphological characterization.

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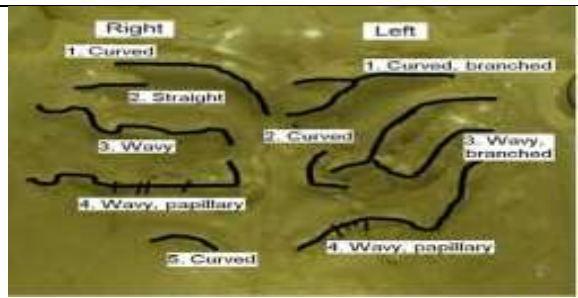


Figure 5 : This image shows the shape Analysis of the palatal Rugae pattern.

3.4. Enhanced Siamese Network Architecture

A Siamese network consists of two identical sub-networks sharing weights. Images A and B are processed in parallel. Their embeddings are compared using a distance metric.

Enhancements Added in Proposed System:

1. ResNet/Inception Backbone instead of simple CNN

2. Multi-layer embedding comparison

3. Contrastive Loss + Triplet Loss hybrid

triplet loss- anchor point, positive point, negative point

Contrastive loss- similar point and dissimilar point

4. Age-invariant feature mapping

ResNet – Residual Network

Better pattern learning with no vanishing gradient and stable training

INCEPTION—

Multiple filter sizes simultaneously to capture features at various scales

this can be used to avoid the age progression

Siamese architecture:

used for the similarity matching between the two images

Similarity Measurement:

$$\text{Euclidean distance: } d = \sqrt{\{(e^1 - e^2)\}} \quad (2)$$

Final decision

$d < \text{threshold} \rightarrow \text{MATCH}$

$d > \text{threshold} \rightarrow \text{NON-MATCH}$

Your enhanced network greatly improves verification accuracy and lowers EER.

3.5. Age Progression Compensation

This is another new contribution in your project. The palatal rugae patterns remain stable but may show minor soft-tissue shifts due to growth. The age compensation module ensures age-invariant recognition.

Methods used:

Temporal feature mapping

Ridge stability analysis

Morphological drift compensation

CNN-based age normalization layer

This module improves recognition for long-term verification, forensic databases, and missing-person tracking.

4.CONCLUSION

The proposed palatal rugae-based human identification system successfully demonstrates that intra-oral biometrics can serve as a highly reliable and scientifically robust alternative to traditional biometric modalities. By integrating advanced preprocessing techniques such as CLAHE, median filtering, ROI extraction, image normalization, and resizing, the system ensures that the input data is uniformly enhanced and noise-free. This preprocessing pipeline significantly improves the clarity and consistency of palatal features, which are essential for accurate deep-learning-based recognition.

The core of the system—the ResNet–Inception–based Siamese twin encoder model—effectively learns the subtle ridge patterns, bifurcations, shapes, and curvatures that distinguish one individual from another. Unlike classical CNN approaches, the hybrid architecture captures multi-scale texture information and deep structural features, enabling stable and discriminative embedding generation. During testing, similarity is determined using the Distance Parameter (DP), ensuring precise matching even with limited training data. The system achieves high robustness against illumination variations, partial distortions, and minor positional changes, which commonly affect oral images.

Overall, the project provides a novel, automated, and highly dependable methodology for human identification using palatal rugae. It bridges an important gap in forensic science and biometric research by offering a secure, tamper-resistant, and biologically stable identification technique. The results confirm that palatal rugae, when analyzed using modern deep learning, can

serve as a powerful and scalable biometric trait suitable for real-world applications such as disaster victim identification, forensic investigations, dental record verification, and secure authentication systems.

5. FUTURE ENHANCEMENT

Although the proposed system performs effectively, several enhancements can further improve its accuracy, scalability, and real-world integration. One major direction is the development of real-time intra-oral scanning using mobile or IoT-enabled dental devices, enabling quick and contactless acquisition of palatal images. This would reduce dependency on static photographs and make the technology more user-friendly in field applications.

Another improvement is the integration of 3D palatal rugae modeling, using depth sensors or structured-light imaging. A 3D approach would capture additional geometric depth variations that are not fully represented in 2D images, offering higher accuracy and robustness. Combining 2D and 3D data could create a multimodal hybrid system that surpasses the limitations of each technique individually.

In the computational aspect, future work could explore Transformer-based encoders, Vision Transformers (ViT), or hybrid CNN-Transformer models to extract richer representations of the palatal region. These models can capture long-range spatial dependencies more effectively, further refining the similarity learning process. Additionally, incorporating self-supervised learning may reduce the need for labeled data, making the system scalable to large populations with minimal manual annotation.

From an application standpoint, the system can be expanded to include multi-biometric fusion, combining palatal rugae with dental pattern analysis, tongue print biometrics, or oral cavity geometry. Such fusion would significantly increase identification confidence in forensic and security applications. Cloud-based storage and secure biometric databases can also be implemented to support large-scale deployment across forensic labs, hospitals, and law enforcement agencies.

Finally, developing a user-friendly mobile application or web interface with secure access control would make the system practical for real-time identification in emergency scenarios. Continuous training with larger and more diverse datasets will enhance generalization and enable the system to perform reliably across different ages, populations, and imaging conditions.

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